Data network backbone Specification

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Revision	C
Status	APPROVED
Classification	INTERNAL
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1. Purpose

This specification defines the performance, interface and verification requirements for the data network backbone (CN-DNB) of the Computer Core & Data Network on Starhaven Orbital Habitat & Shipyard.

2. Scope

Fabrication and installation are performed under work package WP-SHV-CN-03 (WBS 1.12.3), owned by Ursel Marrek.

The prime supplier for flight hardware is SoroSuub Corporation. Responsible engineer: Vira Renner.

3. Applicable documents

Document	Title
SHV-SYS-SPEC-0001	Starhaven Master System Specification
SHV-CN-DWG-0202	Data network backbone general arrangement drawing
SHV-CN-BOM-0302	Data network backbone bill of materials
SHV-QA-PLN-0004	Starhaven Verification and Test Plan
SHV-QA-STD-0002	Station Workmanship Standard
SHV-ICD-HR-CN	Interface Control Document: Hypermatter Reactor / Computer Core & Data Network
SHV-ICD-HB-CN	Interface Control Document: Docking & Hangar Bays / Computer Core & Data Network
SHV-ICD-CN-CM	Interface Control Document: Computer Core & Data Network / Communications Relay
SHV-ICD-CN-OC	Interface Control Document: Computer Core & Data Network / Operations Centre
SHV-ICD-CN-TR	Interface Control Document: Computer Core & Data Network / Internal Transit
SHV-ICD-RL-CN	Interface Control Document: Research Laboratories / Computer Core & Data Network

4. Description

The computer core and data network connect every station system. Maintenance terminals and data ports give technicians and service droids access to the network.

The data network backbone is delivered as a set of line-replaceable assemblies, integrated on station by Construction Operations and verified by Quality Assurance & Test.

5. Performance requirements

ID	Requirement	Value	Tolerance	Verification
CN-DNB-R01	The data network backbone shall provide a backbone links of 234 ea.	234 ea	±2%	Test
CN-DNB-R02	The data network backbone shall not exceed a latency of 3.6 ms.	3.6 ms	±1%	Demonstration

5.1 General requirements

ID	Requirement	Verification
CN-DNB-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.	Inspection
CN-DNB-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.	Analysis
CN-DNB-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.	Inspection
CN-DNB-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.	Analysis
CN-DNB-G05	Documentation: Acceptance data packages shall be delivered with every unit.	Inspection

6. Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7. Verification

Each requirement is verified by the method stated in section 5, following SHV-QA-PLN-0004. Acceptance test reports are filed under the CN test-report series.

8. Revision history

Rev	Date	Description	Author
A	2023-10-10	Initial release.	Vira Renner
B	2024-02-06	Updated requirement values after design review.	Vira Renner
C	2024-08-09	Clarified verification methods.	Vira Renner